hdoc day 38 by trishil chauhan

Aim

To create a menu-driven C program using separate functions to count spaces, digits and special characters, find the frequency of a given character, and toggle the case of letters in a string.

C Program

```
#include <stdio.h>

#define MAX 500

void readString(char str[]) {
    printf("Enter a string: ");
    fgets(str, MAX, stdin);
    int i = 0;
    while (str[i] != '\0') {
        if (str[i] == '\n') {
            str[i] = '\0';
            break;
        }
        i++;
    }
}

void countTypes(const char str[]) {
    int spaces = 0, digits = 0, special = 0;
    for (int i = 0; str[i] != '\0'; i++) {
        char ch = str[i];
        if (ch == ' ')
            spaces++;
        else if (ch >= '0' && ch <= '9')
            digits++;
        else if (!((ch >= 'A' && ch <= 'Z') ||
                   (ch >= 'a' && ch <= 'z')))
            special++;
    }
    printf("Spaces: %d\nDigits: %d\nSpecial characters: %d\n",
           spaces, digits, special);
}

void characterFrequency(const char str[]) {
    char ch;
    int count = 0;
    printf("Enter character to count: ");
    scanf("%c", &ch);
    while (getchar() != '\n');
    for (int i = 0; str[i] != '\0'; i++) {
        if (str[i] == ch)
            count++;
    }
    printf("Frequency of '%c': %d\n", ch, count);
}

void toggleCase(char str[]) {
    for (int i = 0; str[i] != '\0'; i++) {
        if (str[i] >= 'a' && str[i] <= 'z')
            str[i] -= ('a' - 'A');
        else if (str[i] >= 'A' && str[i] <= 'Z')
            str[i] += ('a' - 'A');
    }
    printf("Toggled string: %s\n", str);
```

```
}

int main(void) {
    char str[MAX];
    int choice;
    readString(str);
    do {
        printf("\n--- STRING MENU ---\n");
        printf("1. Count spaces, digits and special characters\n");
        printf("2. Count frequency of a character\n");
        printf("3. Toggle case of each character\n");
        printf("4. Exit\nEnter your choice: ");
        if (scanf("%d", &choice) != 1) {
            printf("Invalid input.\n");
            break;
        }
        while (getchar() != '\n');
        switch (choice) {
            case 1: countTypes(str); break;
            case 2: characterFrequency(str); break;
            case 3: toggleCase(str); break;
            case 4: printf("Exiting program.\n"); break;
            default: printf("Invalid choice. Try again.\n");
        }
    } while (choice != 4);
    return 0;
}
```

Sample Output

```
Enter a string: Hello 123! World

--- STRING MENU ---
Enter your choice: 1
Spaces: 2
Digits: 3
Special characters: 1

Enter your choice: 2
Enter character to count: l
Frequency of 'l': 3

Enter your choice: 3
Toggled string: hELLO 123! wORLD

Enter your choice: 4
Exiting program.
```

Result

The program performs all three string operations using separate functions and exits when option 4 is selected. Spaces are counted separately; special characters include punctuation and other non-alphanumeric, non-space characters.